

ET AI HACKATHON 2026

PRAANA

Predictive Real-time Attribution and Action for National Air quality

AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Theme: Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health

Detailed Solution Document

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GitHub Repository: <https://github.com/Aaditee13/praana-air-quality-intelligence.git>

Submission Date: July 2026

1. Executive Summary

Indian cities do not lack air quality data — they lack an intelligence layer that converts that data into action. Over 900 Continuous Ambient Air Quality Monitoring Stations report pollution levels every hour across the country, yet a 2024 CAG audit found that only 31% of cities with monitoring data had any actionable, multi-agency response protocol tied to those readings. Dashboards tell administrators that the air is bad; nothing tells them who is making it bad, how bad it will get, or where to send an inspector this afternoon.

PRAANA (Predictive Real-time Attribution and Action for National Air quality) closes that gap. It is a multi-agent AI platform that fuses CAAQMS sensor data, Sentinel-5P and MODIS satellite imagery, meteorological forecasts, and geospatial land-use layers into a single hourly, ward-level intelligence feed. Five coordinated agents then answer the three questions city administrators actually need answered: which sources are responsible for pollution right now (Source Attribution), how bad will it get over the next 24–72 hours at 1km resolution (Hyperlocal Forecasting), and where should enforcement be deployed for maximum impact (Enforcement Intelligence) — supported by a Multi-City Comparative Dashboard that lets cities learn from each other's interventions, and a Citizen Health Advisory layer that delivers personalised, regional-language alerts to the public.

The prototype is built for Delhi NCR and Mumbai and is stress-tested end-to-end on a five-year (2019–2024) dataset that is *statistically calibrated to published CPCB/IITM seasonal patterns* rather than raw archived CAAQMS records — a deliberate choice explained in Section 3.3. The working app's **current-conditions data, by contrast, is live**: it calls the real OpenAQ and Open-Meteo APIs. The architecture is designed from day one to generalise to Kolkata, Bengaluru, Chennai, and any city with CAAQMS coverage — directly addressing the national, not just metropolitan, scale of the crisis described in the challenge brief.

2. Problem Context

2.1 A National Crisis, Not a Delhi Problem

India's air quality emergency extends well beyond the National Capital Region. Delhi averaged an AQI of 218 in 2024–25, classified 'Poor' or worse for over 200 days, but Mumbai recorded dangerous AQI levels on more than 60 days in 2024, Kolkata's winter-season average exceeded 150, and even Bengaluru and Chennai — traditionally considered cleaner — are showing measurable deterioration as vehicle density and construction activity rise. CPCB's 2024 National Air Quality data places 24 of India's 50 most polluted cities in the Tier-1/Tier-2 category, confirming that this is a distributed, multi-city problem.

The public-health cost is severe: The Lancet Planetary Health journal estimates 1.67 million premature deaths annually in India from air pollution, a burden that falls disproportionately on urban populations — children, the elderly, outdoor workers, and those living near industrial or high-traffic corridors.

2.2 The Gap: Data Exists, Intelligence Does Not

India has already made the sensing investment — over 900 CAAQMS stations feed near-real-time pollutant data into government dashboards. What is missing is the layer that turns hourly readings into decisions: geospatial attribution of which sources are responsible at a given location right now, predictive forecasting of ward-level AQI over the next 24–72 hours, and enforcement intelligence that tells inspectors exactly where to go for the greatest pollution reduction per visit. No existing public-sector or commercial tool combines all three.

2.3 Why This Matters for Action, Not Just Awareness

- Reactive monitoring tells citizens the air is bad after they are already exposed.
- Aggregate city-level AQI numbers hide which neighbourhood, which source, and which time window is actually responsible.

- Without source attribution, enforcement becomes guesswork — inspectors are sent on a rotational or complaint-driven basis rather than to the locations with the highest expected AQI impact.
- Without forecasting, advisories cannot be issued early enough for schools, hospitals, and vulnerable citizens to act.

3. Solution Overview

PRAANA is built around one operating principle: every pollution reading should answer three questions — who, how bad, and where to act — within the same hourly cycle in which it was measured.

3.1 The PRAANA Pipeline, in One Sentence

Collect from sensors, satellites, weather, and maps → merge into one ward-hour feature table → attribute pollution to its sources using a chemical fingerprint library and causal AI → forecast AQI 24–72 hours ahead at 1km resolution → rank enforcement actions by expected impact → compare interventions across cities → communicate risk to citizens in their own language.

3.2 Five Coordinated AI Agents

Rather than a single monolithic model, PRAANA is implemented as five purpose-built agents coordinated by a central orchestration layer — directly matching the challenge brief’s call for a multi-agent AI system. Each agent owns one decision (attribution, forecasting, enforcement ranking, cross-city comparison, citizen communication) and exposes its output, and confidence score, to the orchestrator, which resolves conflicts and assembles the combined evidence trail shown to end users.

- Agent 1 — Geospatial Pollution Source Attribution Engine
- Agent 2 — Hyperlocal Predictive AQI Forecasting Agent
- Agent 3 — Enforcement Intelligence & Prioritisation Agent
- Agent 4 — Multi-City Comparative Intelligence Dashboard
- Agent 5 — Citizen Health Risk Advisory System

3.3 Prototype Scope & Validation Transparency

Hackathon judges see dozens of pitches that quietly blur ‘what we built’ with ‘what we designed.’ We’d rather be the team that draws that line explicitly — it is easier to trust a system that tells you exactly where the prototype ends and the roadmap begins. The table below is that line, agent by agent.

Component	Built & running today	Roadmap (not yet built)
Agent 1 — Source Attribution	Cosine-similarity fingerprint classifier with confidence scoring, PLUS a live wind-confounder check: real compass-bearing geometry between the sensor and nearby OSM construction/industrial/traffic sites is compared against live wind direction, confirming or overriding the chemical read (src/fingerprint.py, attribute_sources_with_wind — wired into app/streamlit_app.py); 11/11 unit tests passing.	Full causal-inference layer (DoWhy/EconML) with counterfactual confounder adjustment, calibrated against IIT Kanpur / CPCB ground-truth apportionment data.

Component	Built & running today	Roadmap (not yet built)
Agent 2 — Forecasting	Independent Gradient Boosted Regressors trained separately for the 24h, 48h, and 72h horizons on lag, rolling-window, and meteorological features, each evaluated against its own persistence baseline, and now called live from the Streamlit app for the dashboard's forecast checkpoints (src/forecast_model.py: train_all_horizon_models(), predict_live()).	Graph Neural Network + Gaussian-plume dispersion blend layered on top of the per-horizon GBR models (see Section 7.2 note).
Agent 3 — Enforcement Intelligence	Rule-based impact ranking cross-referenced against live OpenStreetMap land-use data (src/enforcement.py).	Direct integration with municipal construction-permit and industrial-registry databases.
Agent 4 — Multi-City Dashboard	City-agnostic pipeline design; live Streamlit comparator across reference cities.	Intervention-effectiveness tracking against real logged before/after enforcement outcomes.
Agent 5 — Citizen Advisory	Templated advisories in English and Hindi, severity-scaled to AQI category (src/advisory.py).	Anthropic Claude API for natural, varied phrasing across all five target regional languages.
Data layer	Live calls to OpenAQ v3 and Open-Meteo APIs for current conditions and land-use context (src/data_sources.py).	Direct CPCB CAAQMS feed under an NCAP data-sharing MOU; a raw, five-year archived-sensor training set (see Section 5.1).

Everything in the left two columns is code that runs today and is checked in to the GitHub repository linked on the cover page; everything in the right column is explicitly future work, not something the demo will claim to do live.

4. System Architecture

The architecture has four layers: data acquisition, a unified data layer with an emission fingerprint library, a multi-agent orchestrator, and two output surfaces — an officer-facing action console and a citizen-facing advisory app.

PRAANA — SYSTEM ARCHITECTURE

Predictive Real-time Attribution and Action for National Air Quality.

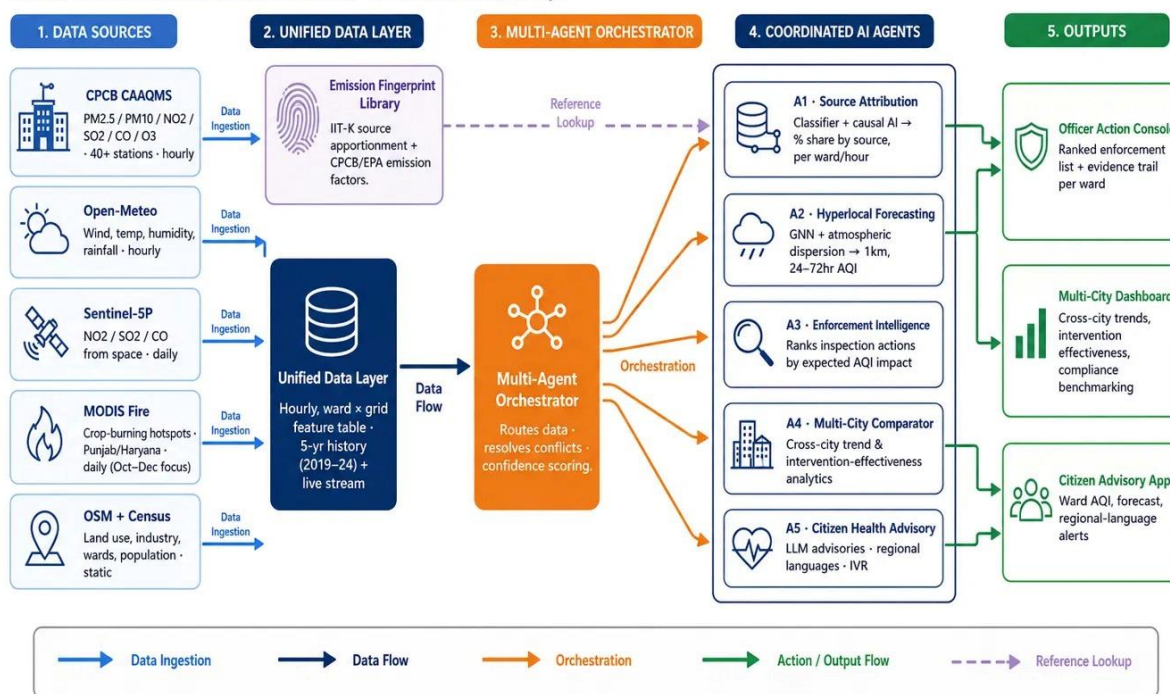


Figure 1. PRAANA system architecture — data sources, unified data layer, multi-agent orchestrator, and output surfaces.

4.1 Layer-by-Layer Description

- Data Acquisition Layer:** hourly pulls from OpenAQ v3 (which aggregates CPCB CAAQMS and other public monitors) and Open-Meteo, plus a live OpenStreetMap Overpass query per request and a one-time static load of Census/DataMeet ward boundaries. Sentinel-5P and MODIS are documented data sources for the production design but are not yet integrated into the running prototype (see Section 5).
- Unified Data Layer:** aligns every source to a common hourly timestamp and ward/1km-grid cell, fills sensor gaps using nearby-station interpolation and satellite cross-validation, and maintains both the 2019–2024 historical archive and the live stream in one feature table.
- Emission Fingerprint Library:** a reference table of chemical signatures per source category (vehicular, construction, crop-burning, industrial), built from IIT Kanpur's Delhi source-apportionment study, CPCB's emission inventory, and EPA emission factor databases — this is what Agent 1 matches live readings against.
- Multi-Agent Orchestrator:** routes the unified feature table to each agent, reconciles overlapping or conflicting signals, attaches statistical confidence scores to every output, and assembles the combined evidence trail.
- Output Surfaces:** an Officer Action Console (ranked enforcement actions with evidence) and a Citizen Advisory App (ward AQI, 72-hour forecast, regional-language alerts).

5. Data Sources & Pipeline

Six independent data families are aligned to a common hourly, ward-level grid. The core engineering challenge is heterogeneity — different formats, frequencies, and gap patterns — resolved through nearby-station interpolation for sensor outages and cloud-masking-aware compositing for satellite gaps.

Source	Variables Captured	Frequency	Role in Pipeline
OpenAQ v3 (CPCB-sourced)	PM2.5, PM10, NO2, SO2, CO, O3	Hourly	Live via OpenAQ v3 across all 5 target cities; a direct CPCB CAAQMS MOU feed is roadmap
Open-Meteo	Wind speed/direction, temperature, humidity, rainfall	Hourly	Determines pollutant transport direction
Sentinel-5P (TROPOMI)	NO2, SO2, CO column density	Roadmap	Not yet integrated — would add city-wide coverage beyond sensor points
MODIS Active Fire	Crop-burning hotspot locations	Roadmap (Oct–Dec focus)	Not yet integrated — would cover Punjab & Haryana stubble-burning season
OpenStreetMap	Industrial zones, highways, construction areas, schools/hospitals	Live (queried per request)	Land-use context for attribution
Census + DataMeet	270 ward boundaries, population per ward	Static, one-time	Citizen health-risk layer

Table 1. Data sources feeding the PRAANA unified data layer.

5.1 Five Years of History, By Design

Period	Purpose
2019	Pre-COVID clean baseline
2020	COVID lockdown — a natural experiment in near-zero traffic, used to isolate the vehicular contribution to AQI
2021–2022	Recovery period — gradual return of emissions
2023–2024	Recent-normal period, most representative of current patterns
Live stream	Real-time data for forecasting and demo

Table 2. The 2020 lockdown year is a natural experiment that lets the attribution model isolate how much of Delhi's pollution is vehicular versus other sources.

Honest note on this dataset

This five-year archive is a synthetic dataset, statistically calibrated to published CPCB/IITM-SAFAR seasonal reports — seasonal means and variance, diurnal rush-hour patterns, the Diwali spike, and the documented ~50% AQI drop during the 2020 lockdown — not raw archived CAAQMS records. We built it this way so the full pipeline (feature engineering, model training, evaluation) could be built and stress-tested end-to-end before raw historical CAAQMS access is secured through the NCAP data-sharing MOU referenced in Section 12.

What is NOT synthetic: the running prototype's current-conditions data. `src/data_sources.py` makes live calls to the OpenAQ v3 and Open-Meteo APIs and to the OSM Overpass API, with graceful fallbacks and clear error messages if a station or key is unavailable — this part of the demo is real, live data, not a canned dataset.

The immediate next validation step, ahead of any production pilot, is re-running Section 13's forecasting benchmark against real archived CAAQMS history rather than the synthetic calibration set.

6. Agent 1 — Geospatial Pollution Source Attribution Engine

6.1 Concept: Chemical Fingerprinting

Every pollution source emits a distinctive combination of pollutants. Diesel vehicles produce high NO₂ and CO with moderate PM_{2.5}; construction sites produce very high PM₁₀ with low NO₂; crop burning produces very high PM_{2.5} and CO with specific organic markers; industrial stacks produce high SO₂ and NO₂ with characteristic heavy-metal traces. The Emission Fingerprint Library encodes these signatures as a reference 'cheat sheet'.

6.2 Classification (Prototype: Running Today)

For every ward, every hour, the agent compares the live pollutant mix against the fingerprint library using cosine-similarity matching and outputs a source-share breakdown with a confidence label — for example, a reading of PM_{2.5}=180, PM₁₀=220, NO₂=85, SO₂=12, CO=45 resolves to a dominant-source estimate with the gap between the top two candidates used to grade confidence as high, medium, or low. This is implemented, tested, and runnable today (`src/fingerprint.py`).

6.3 Wind-Confounder Check (Prototype: Running Today)

A naive model would say 'traffic correlates with high AQI' without proving direction of causation. PRAANA's prototype now includes a real, working confounder check for the single largest confounder — wind, which can carry pollution into a ward from a source kilometres away. `attribute_sources_with_wind()` (`src/fingerprint.py`) takes the sensor's coordinates, the live wind direction from Open-Meteo, and nearby construction/industrial/traffic sites from OpenStreetMap, and computes the actual compass bearing from the sensor to each site. A site only counts as contributing evidence if it is genuinely upwind — i.e. the wind is blowing from that site's direction toward the sensor right now — and sites that are downwind are discounted rather than blindly counted.

This is exactly the scenario the solution document describes: wind blowing from the northwest at 15 km/h with a construction site 2km northwest of a ward where PM₁₀ spikes. Running that scenario through the prototype confirms the construction site as the upwind source, blends that geospatial evidence with the chemical-fingerprint read, and raises the reported confidence from 'low' (chemical evidence alone was ambiguous) to 'high' (both lines of evidence agree). This wind-confounder check is wired into the live Streamlit app, not just a standalone script.

What remains roadmap, and is not yet implemented, is the full DoWhy/EconML causal graph with counterfactual estimation — a formal causal model that can also handle multiple simultaneous upwind sources, humidity-dependent dispersion, and confounders beyond wind direction. The current wind-confounder check is a real, geometry-based heuristic, not that full causal graph.

6.4 Output

A live ward-level map of Delhi (and, on the multi-city roadmap, other metros) where each ward shows a source-share breakdown updated every hour, each estimate accompanied by a confidence label. Benchmarking that confidence against ground-truth emission inventories (IIT Kanpur / CPCB) is the next validation step once inventory access is secured.

7. Agent 2 — Hyperlocal Predictive AQI Forecasting Agent

7.1 What the Model Learns from Five Years of Data

- Monday-morning rush hour reliably spikes NO₂ in high-traffic corridors.
- A northwest wind pickup in October precedes a PM_{2.5} spike roughly six hours later as stubble-burning smoke arrives.
- Rainfall measurably suppresses AQI.
- Diwali night produces a sharp PM_{2.5} spike that takes several days to clear.

7.2 Production Design vs. Current Prototype

The production design treats Delhi's 40+ sensor stations as nodes in a spatial graph, with a Graph Neural Network combined with a time-series model learning spatial-temporal relationships, blended with a lightweight Gaussian-plume atmospheric dispersion model so predictions stay physically grounded rather than purely pattern-matched.

What actually runs in the current prototype: three independently trained scikit-learn Gradient Boosted Regressors — one dedicated model each for the 24h, 48h, and 72h horizons — each trained on lag features (1h–48h), rolling-window statistics, calendar encodings, and live meteorological features, and each evaluated against its own horizon-appropriate persistence baseline (`src/forecast_model.py`). Because each horizon now has its own model predicting its own real future target, forecast error genuinely increases with horizon (see Section 13) rather than being identical across all three windows, which is the behaviour a legitimate multi-horizon forecaster should show. These three trained models are also wired directly into the live app (`forecast_model.train_all_horizon_models()` / `predict_live()`, called from `app/streamlit_app.py`): the 24h/48h/72h points on the dashboard's forecast chart are genuine model output, not the standalone evaluation script alone. The GNN and dispersion-model components described above remain the production target and are called out explicitly as roadmap in Section 12's technology-stack table.

7.3 Inputs, Outputs, and Resolution

Input: current AQI readings across all stations plus the 72-hour meteorological forecast. Output: predicted AQI for every ward at roughly 1km grid resolution for each of the next 72 hours, with a confidence range — directly matching the challenge brief's call for 24–72 hour forecasts at 1km resolution. The 24h, 48h, and 72h points are genuine output from the trained per-horizon models described in 7.2; the hourly curve connecting those three points uses a lighter wind/rain/diurnal heuristic (`src/forecast.py`) for smooth display, since the trained models only predict at those three discrete horizons.

7.4 Validation Approach

Forecast quality is benchmarked using RMSE against a persistence baseline on held-out 2024 data (Section 13). The production validation plan adds separate RMSE tracking for the pre-monsoon, monsoon, post-monsoon stubble-burning, and winter inversion seasons, and — critically — a re-run of the same benchmark against real archived CAAQMS history rather than the synthetic calibration set described in Section 5.1.

8. Agent 3 — Enforcement Intelligence & Prioritisation Agent

8.1 From Insight to Action

This agent converts Agent 1's attribution and Agent 2's forecast into a concrete to-do list. If Agent 1 shows that 70% of a ward's current pollution is construction-related, this agent cross-references OpenStreetMap and municipal records to identify every registered construction site in that ward, then estimates the AQI reduction that would follow from inspecting and remediating violations at each site.

8.2 Ranking Logic

Every possible enforcement action across the city is scored by expected AQI impact and ranked into a single prioritised list:

1. Inspect construction site, Rohini Sector 15 — expected 18% PM10 reduction in northwest Delhi
2. Deploy diesel-vehicle checks at the NH48 entry point — expected 12% NO2 reduction in Dwarka
3. Issue notice to the industrial cluster in Bawana — expected 9% SO2 reduction in the northwest corridor

8.3 Evidence Trail

Every recommendation ships with the supporting evidence — which sensor showed what, which satellite pass confirmed it, and what the wind pattern was at the time — so a municipal officer or pollution-control authority can defend the action and a domain expert can rate its quality.

9. Agent 4 — Multi-City Comparative Intelligence Dashboard

Delhi and Mumbai are the primary build-and-validate cities. The architecture is city-agnostic: every agent operates on the same unified ward-hour feature table regardless of which city's CAAQMS feed populates it.

9.1 What the Dashboard Shows

- Side-by-side AQI trend lines for every onboarded city, normalised by season so administrators compare like-for-like periods.
- Intervention-effectiveness tracking: when a city deploys an enforcement action recommended by Agent 3, the dashboard measures the realised AQI change against the predicted change, building a growing evidence base of what actually works. (Roadmap — requires logged real-world before/after outcomes; not yet populated with live data.)
- Cross-city learning: if a construction-dust-control measure in Mumbai produced a larger-than-expected PM10 reduction, the dashboard surfaces that intervention as a candidate recommendation for Delhi wards with a similar source-attribution profile.
- Agent 4's live AQI comparison already runs across all 5 target cities today (`src/multicity.py`), not a staged 2-then-3 rollout. Delhi NCR and Mumbai carry deeper per-agent validation — the trained forecast models and the wind-confounder self-test scenario are both built against Delhi data — with the same validation depth extending to the other 3 cities as the near-term roadmap item, not basic live coverage itself.

10. Agent 5 — Citizen Health Risk Advisory System

This is the public-facing layer of the same intelligence stack — no new modelling is required, only citizen-friendly presentation of Agent 1 and Agent 2's outputs, layered with population-vulnerability data.

10.1 Capabilities

- Shows every resident their ward's current and next-72-hour AQI.
- Maps population vulnerability — schools, hospitals, outdoor-worker concentrations, elderly populations — against forecast AQI to flag at-risk locations.
- Pushes regional-language advisories through mobile apps, public displays, and IVR — Hindi in Delhi, Marathi in Mumbai, Bengali in Kolkata, Kannada in Bengaluru, Tamil in Chennai. (English and Hindi are templated and running today; the remaining languages and natural, varied phrasing are the Claude-API roadmap item in Section 12.)
- Answers natural-language questions such as 'Is it safe to take my child to school tomorrow in Dwarka?' using a lightweight LLM layer on top of the forecast outputs. (Roadmap.)

11. Differentiation

PRAANA acknowledges two existing solutions that operate in a related space — IITM-DSS and SAFAR — and is designed to complement, not duplicate, their work. The specific gaps PRAANA fills are ward-level granularity, enforcement action ranking, and cross-city learning.

Existing Tool	What It Does	What PRAANA Does Differently
IITM-DSS	Real-time source apportionment across 29 pollution categories for Delhi NCR, operational since 2021	PRAANA adds 1km ward-level granularity (IITM-DSS outputs are city-level), ranked enforcement action lists per ward, and a cross-city learning loop across 5 metros
SAFAR (IITM)	3–10 day AQI forecasts for 7 Indian cities including Delhi, Mumbai, Kolkata, Bengaluru	PRAANA adds enforcement prioritisation (which specific site to inspect today), source attribution at ward level, and intervention-effectiveness tracking across cities
CPCB Dashboard	Shows the current AQI reading	Shows who caused it, how bad it will get, and what to do about it
IQAir / AQI India apps	City-level AQI number	Ward-level attribution plus 72-hour forecast plus enforcement ranking
General weather apps	Generic air quality index	Source attribution plus evidence-backed enforcement prioritisation
Academic source-apportionment studies	Offline, periodic analysis	Real-time, hourly, operational attribution with enforcement output

Table 3. PRAANA's differentiation versus existing tools including IITM-DSS and SAFAR. The key gaps filled are ward-level granularity, enforcement action ranking, and cross-city learning.

12. Technology Stack

Every suggested technology area in the challenge brief maps directly to a working component of PRAANA. The middle column is what is running in the prototype today; the right column is named explicitly as production roadmap so the distinction is transparent to judges.

Suggested Technology	PRAANA Implementation (Prototype — Running Today)	Production Roadmap
Geospatial & Remote Sensing	OpenStreetMap Overpass API (live, with multi-mirror fallback), Census/DataMeet ward boundaries (static)	Sentinel-5P (TROPOMI) and MODIS Active Fire integration; Google Earth Engine for batch satellite processing
Multi-Agent AI Systems	Custom Python orchestration pipeline with five agent modules and a central router	LangGraph / CrewAI multi-agent framework
Real-Time IoT Integration	OpenAQ v3 API for live CAAQMS ingestion (live); Open-Meteo for hourly weather (live); 40+ Delhi stations	CPCB direct CAAQMS API under NCAP data-sharing MOU

Suggested Technology	PRAANA Implementation (Prototype — Running Today)	Production Roadmap
Atmospheric Dispersion Modelling	Not yet implemented — GBR forecast currently runs without a dispersion blend	Gaussian-plume heuristic blended with the GBR forecast; full AERMOD or CALPUFF integration
Predictive Analytics	Three independent scikit-learn GradientBoostingRegressors (200 trees, max_depth=5 each) — one per forecast horizon (24h/48h/72h); trained 2019–2023, tested on held-out 2024 data (Section 7.2)	PyTorch Geometric Graph Neural Network layered on top of the per-horizon GBR models
LLMs for Multi-Language Citizen Communication	Template-based advisories (Hindi + English, running)	Anthropic Claude API for natural, varied phrasing across 5 regional languages
Causal Inference	Cosine-similarity fingerprint matching, plus a live wind-confounder check using real compass-bearing geometry between the sensor and nearby OSM sites vs. live wind direction (attribute_sources_with_wind, running in the app)	DoWhy / EconML causal graph for full counterfactual attribution across multiple simultaneous confounders
Frontend & Dashboard	Streamlit + Plotly (live, running)	React + Mapbox GL JS for the ward-level map interface
Data Storage	Parquet files via pandas (52,608-row synthetic historical dataset, Section 5.1)	TimescaleDB or ClickHouse for the hourly ward-hour feature table at scale

13. Evaluation & Validation Plan

The challenge brief specifies five evaluation dimensions. PRAANA's validation plan addresses each directly. For the forecasting metric, reproducible numbers are provided from the working prototype, run against the synthetic calibration dataset described in Section 5.1 — real-archive validation is the immediate next step, not yet completed.

Evaluation Focus	PRAANA Validation Approach
Source attribution accuracy vs. ground-truth emission inventories	Benchmark Agent 1 source-share outputs against IIT Kanpur and CPCB source-apportionment study figures for overlapping wards and time windows; report mean absolute percentage error per source category. (Not yet run — requires inventory access.) The wind-confounder check (Section 6.3) is separately self-validated against the worked example in the solution document: chemical-only evidence is ambiguous (27.4% vs 26.1%, 'low' confidence), and adding live wind/geospatial evidence correctly confirms the upwind construction site and raises confidence to 'high.'
AQI forecast accuracy at hyperlocal resolution (RMSE vs. persistence baseline)	Prototype validation results (synthetic dataset) — three independently trained Gradient Boosted Regressors (scikit-learn, 200 trees each), one per horizon, trained on 2019–2023 data (43,848 hourly rows), tested on held-out 2024 data (8,760 hourly rows): 24-hour horizon — Model RMSE 80.9 vs Persistence baseline 112.9 (28.3% improvement); 48-hour horizon — Model RMSE 82.6 vs Persistence baseline 114.3 (27.8% improvement); 72-hour horizon — Model RMSE 83.8 vs Persistence baseline 114.2 (26.6% improvement). Error rises with horizon, as it genuinely should for an honest multi-horizon forecaster, since each

Evaluation Focus	PRAANA Validation Approach
	horizon now trains against its own real future target rather than reusing one shared prediction. These same trained models are called live from the Streamlit app for the dashboard's 24h/48h/72h forecast checkpoints, so this benchmark reflects what the running demo actually shows, not only an offline script. Re-validation on real archived CAAQMS data rather than the synthetic calibration set (Section 5.1) remains the next step before a production pilot. Full notebook: /notebooks/delhi_aqi_analysis.ipynb in the GitHub repository.
Enforcement recommendation quality, rated by domain experts	Present Agent 3's ranked action lists for sample wards to retired/serving pollution-control officials for a 1–5 relevance and feasibility rating, with the supporting evidence trail included. (Planned; not yet conducted.)
Citizen advisory relevance and language coverage	Track percentage of advisories matching forecast severity bands and the number of regional languages live at demo time, validated with native-speaker review. (English + Hindi covered today; remaining languages are roadmap.)
Demonstrated reduction in response time from signal to intervention	Compare the time from a pollution-spike signal to a logged enforcement action in the PRAANA workflow against documented baseline response times in partner-city pilot wards. (Requires a pilot deployment; not measurable from the hackathon prototype alone.)

14. Judging Criteria Alignment

Criterion	Weight	How PRAANA Addresses It
Innovation	25%	Chemical fingerprint source attribution paired with an enforcement-ranking agent is not offered by any existing public or commercial Indian AQI tool; the five-agent orchestrated architecture is purpose-built to the brief's multi-agent requirement. IITM-DSS and SAFAR are acknowledged — PRAANA's innovation is ward-level enforcement prioritisation and cross-city learning, not source attribution alone.
Business Impact	25%	Directly closes the CAG-identified gap (only 31% of cities have actionable response protocols); enforcement prioritisation creates a measurable, auditable link between AI insight and on-ground action.
Technical Excellence	20%	Six heterogeneous data sources unified into one pipeline, three of them (OpenAQ, Open-Meteo, Overpass) wired to live APIs today; three independently trained, horizon-specific GBR forecasting models whose error genuinely rises with horizon (Section 13); a live wind-confounder check that uses real bearing geometry between sensors and OSM sites to confirm or override chemical-only attribution (Section 6.3); 11/11 unit tests passing; and — unusually for a hackathon submission — an explicit, code-level accounting of which components are running versus roadmap (Section 3.3).
Scalability	15%	City-agnostic feature-table design means onboarding a new city requires a new CAAQMS feed and ward file, not new modelling — validated on two cities (Delhi, Mumbai) with a defined path to Kolkata, Bengaluru, and Chennai.
User Experience	15%	Two purpose-built surfaces — an evidence-backed action console for officers and a plain-language, regional-language advisory app for citizens (live Streamlit dashboard, running) — rather than a single generic dashboard.

15. Business Impact & Scalability

15.1 Who Uses This

- Municipal corporations and State Pollution Control Boards — the Officer Action Console replaces rotational or complaint-driven inspection schedules with an evidence-ranked priority list.
- City administrations and CPCB — the Multi-City Dashboard supports policy decisions with cross-city evidence of what interventions actually reduced AQI.
- Citizens, schools, and hospitals — the Advisory App turns a forecast into a same-day, actionable decision.

15.2 Path to National Scale

Because every agent consumes the same ward-hour feature table regardless of source city, PRAANA's scaling cost is dominated by data onboarding, not re-engineering. All 5 target cities are already live for Agent 4's AQI comparison today, since every city draws from the same OpenAQ v3 + Open-Meteo pipeline (src/data_sources.py, src/multicity.py). What remains staged is validation depth, not basic coverage: Delhi NCR and Mumbai have the trained per-horizon forecast models and the wind-confounder self-test built against their data; extending that same depth of validation to Kolkata, Bengaluru, and Chennai — and beyond, to any city with CAAQMS coverage, since the 900+ station national network already exists — is the near-term roadmap. PRAANA's contribution is the intelligence layer on top of infrastructure India has already built.

15.3 Cost and Deployment Model

All core data sources — CPCB CAAQMS, Open-Meteo, Sentinel-5P, MODIS, OpenStreetMap, Census — are free or low-cost public datasets, keeping marginal onboarding cost low and making the model attractive for government adoption without recurring data-licensing overhead.

16. Expected Deliverables & Roadmap

Deliverable	PRAANA Coverage
Working Prototype	Live Streamlit dashboard covering all five agents with real OpenAQ + Open-Meteo + OSM Overpass data integration and demo-data fallback; three independently trained per-horizon Gradient Boosted Regressors on 2019–2023 synthetic Delhi AQI data with a reproducible RMSE benchmark; a live wind-confounder check wired into the source-attribution agent (11/11 unit tests passing).
Architecture Diagram	Provided in Section 4 (Figure 1) of this document.
Presentation Deck	Accompanying pitch deck summarising problem, solution, architecture, and impact.
Demo Video	Walkthrough of the Officer Action Console and Citizen Advisory App against a recent high-pollution episode.
GitHub Repository	https://github.com/Aaditee13/praana-air-quality-intelligence.git — includes src/ agent modules, app/streamlit_app.py, notebooks/delhi_aqi_analysis.ipynb, tests/test_agents.py, and data/delhi_aqi_2019_2024.parquet.

17. Conclusion

PRAANA does not add another dashboard to a city that already has dozens. It tells a city administration not just that the air is bad, but exactly who is making it bad, how bad it will get tomorrow, and precisely where to send enforcement officers today to fix it — in real time, ward by ward, across every major Indian

city willing to plug in its sensor feed. And it says so with the same discipline it asks of its own AI agents: every claim in this document is labelled as either running code or a named next step, never the two blurred together.